A PAIRWISE LEARNING-TO-RANK MODEL FOR FEATURE ENGINEERING, LOGISTIC OBJECTIVES, AND CALIBRATION

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ABSTRACT. We develop an interpretable learning-to-rank pipeline for first-round admissions screening in Pi Mu Epsilon, using limited human pairwise comparisons [1, 2, 3] within the pairwise learning-to-rank framework [4, 5, 6, 7]. A deterministic map $x : \mathscr{A} \to [0,10]^{21}$ encodes structured fields, resume text, and short-answer signals as bounded named features [8, Chapter 3]. The linear score $s_w(a) := \langle w, x(a)\rangle$ minimizes the regularized logistic risk [5, 8]

$$L_\lambda(w) := \frac{1}{|\mathscr{D}|} \sum_{(i,j,y_{ij})\in\mathscr{D}} \log\bigl(1+\exp\bigl(-z_{ij}\langle w, x(a_i)-x(a_j)\rangle\bigr)\bigr) + \frac{\lambda}{2}\|w\|_2^2, \qquad z_{ij} := 2y_{ij}-1.$$

For $\lambda > 0$, this objective is λ-strongly convex and $(\lambda + 25\cdot 21)$-smooth; hence gradient descent converges linearly to its unique minimizer [9, Chapter 3]. Raw scores are calibrated by

$$\widehat{s}(a) := 10\,\widehat{F}_w\bigl(s_w(a)\bigr),$$

which maps them to $[0,10]$, preserves order, and is invariant under strictly increasing reparametrization [7], [8, Chapter 4]. Finally, we specify the reproducible data artifacts `applicants.jsonl`, `pairs.csv`, and `rank_model.json` linking local training to deployed scoring.

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Part 1. Pairwise Ranking

1. Introduction

First-round admissions screening produces a consistent and auditable ordering that allocates reviewer attention rather than a final decision, and in the regime of $n \approx 10$–50 applicants per cycle the only reliably collectable supervision is the comparison $a_i \succ a_j$ between two applications [1], [4], [5], [7]. Throughout, $\mathscr{A}$ is the space of applications, $d \in \mathbb{Z}_{\geq 1}$ the feature dimension, $\langle \cdot, \cdot \rangle$ the Euclidean inner product on $\mathbb{R}^d$ with norm $\|\cdot\|_2$, $\mathbf{1}\{\cdot\}$ an indicator, I the identity matrix, $\preceq$ the Loewner order on symmetric matrices, and

$$\sigma(t) := \frac{1}{1+e^{-t}} \quad (t \in \mathbb{R}), \qquad \sigma(-t) = 1 - \sigma(t), \qquad \sigma'(t) = \sigma(t)\,\sigma(-t).$$

1.1. Setting and result.

Definition 1.1.1 (Features, scores, and margins)**.** A feature map is a deterministic function $x : \mathscr{A} \to [0,10]^d$, $a \mapsto x(a) = (x_1(a), \ldots, x_d(a))$, carrying a schema identifier fv $\in \mathbb{Z}_{\geq 1}$. For $w \in \mathbb{R}^d$ and applications a, a_i, a_j,

$$s_w(a) := \langle w, x(a) \rangle = \sum_{k=1}^{d} w_k x_k(a), \qquad d_{ij} := x(a_i) - x(a_j), \qquad \Delta_{ij}(w) := s_w(a_i) - s_w(a_j) = \langle w, d_{ij} \rangle.$$

Definition 1.1.2 (Pairwise dataset and regularized pairwise logistic risk, [1, 5])**.** A pairwise dataset is a finite multiset $\mathscr{D} = \{(i, j, y_{ij})\}$ of pairs of applicant indices with labels $y_{ij} \in \{0,1\}$, $y_{ij} = 1$ meaning $a_i \succ a_j$, and $z_{ij} := 2y_{ij} - 1 \in \{-1,+1\}$. The Bradley–Terry probability, the pairwise logistic loss, and the ℓ_2-regularized risk are

$$\mathbb{P}(a_i \succ a_j \mid w) := \sigma\big(\Delta_{ij}(w)\big), \qquad \ell_{ij}(w) := \log\Big(1 + \exp\big(-z_{ij}\,\Delta_{ij}(w)\big)\Big),$$

$$L_\lambda(w) := \frac{1}{|\mathscr{D}|} \sum_{(i,j,y) \in \mathscr{D}} \ell_{ij}(w) + \frac{\lambda}{2}\|w\|_2^2 \quad (\lambda \geq 0), \qquad \Lambda := \lambda + \frac{1}{4|\mathscr{D}|} \sum_{(i,j,y) \in \mathscr{D}} \|d_{ij}\|_2^2.$$

Theorem A (Regularized pairwise logistic ranking)**.** *Let $x : \mathscr{A} \to [0,10]^d$ be a feature map, $\mathscr{D}$ a pairwise dataset, $\lambda > 0$, $w^{(t+1)} := w^{(t)} - \Lambda^{-1} \nabla L_\lambda(w^{(t)})$, and $\widehat{s}$ the calibrated score of Definition 6.1.1 on a pool $\mathscr{P}$. Then L_λ has a unique minimizer w_λ, $\Lambda \leq \lambda + 25d$, and for all $w \in \mathbb{R}^d$, $t \geq 0$, and $a, b \in \mathscr{P}$,*

$$\lambda I \preceq \nabla^2 L_\lambda(w) \preceq \Lambda I, \qquad s_w(a) \leq s_w(b) \iff \widehat{s}(a) \leq \widehat{s}(b),$$

$$L_\lambda(w^{(t)}) - L_\lambda(w_\lambda) \leq \Big(1 - \frac{\lambda}{\Lambda}\Big)^t \big(L_\lambda(w^{(0)}) - L_\lambda(w_\lambda)\big).$$

Proof. §7 □

2. Preliminaries

2.1. Ranking models, the logistic function, and linear bounds.

Definition 2.1.1 (Bradley–Terry and Plackett–Luce models, [1, 2, 3])**.** For $\theta \in \mathbb{R}^n$ and a permutation π of $\{1, \ldots, n\}$,

$$\mathbb{P}_\theta(i \succ j) := \frac{e^{\theta_i}}{e^{\theta_i} + e^{\theta_j}} = \sigma(\theta_i - \theta_j), \qquad \mathbb{P}_\theta(\pi) := \prod_{k=1}^{n} \frac{e^{\theta_{\pi(k)}}}{\sum_{l=k}^{n} e^{\theta_{\pi(l)}}},$$

the second reducing to the first for $n = 2$.

Fact 2.1.2 (Logistic identities, [8])**.** *For $t \in \mathbb{R}$,*

$$\sigma(-t) = 1 - \sigma(t), \qquad \sigma'(t) = \sigma(t)\,\sigma(-t) \in \big(0, \tfrac{1}{4}\big], \qquad -\log \sigma(t) = \log\big(1 + e^{-t}\big),$$

$$\frac{d}{dt}\log\big(1 + e^{t}\big) = \sigma(t), \qquad \frac{d^2}{dt^2}\log\big(1 + e^{t}\big) = \sigma'(t) > 0, \qquad \frac{d^2}{dt^2}\log\big(1 + e^{-t}\big) = \sigma'(-t) > 0.$$

Lemma 2.1.3 (Linear functionals on a box)**.** *For $w, v \in \mathbb{R}^d$ with $\|v\|_\infty \le 10$,*

$$|\langle w,v\rangle| \le \|w\|_1\|v\|_\infty \le 10\,\|w\|_1, \qquad \|v\|_2^2 \le d\,\|v\|_\infty^2 \le 100\,d, \qquad v,w\in[0,\infty)^d \implies 0\le\langle w,v\rangle\le 10\|w\|_1.$$

Proof.

$$|\langle w,v\rangle| \le \sum_{k=1}^{d}|w_k|\,|v_k| \le \|v\|_\infty\|w\|_1, \qquad \|v\|_2^2=\sum_{k=1}^{d} v_k^2 \le d\|v\|_\infty^2, \qquad v_k,w_k\ge 0 \implies 0\le\langle w,v\rangle\le 10\|w\|_1.$$

□

Lemma 2.1.4 (Level sets of a linear functional)**.** *For $w\in\mathbb{R}^d\setminus\{0\}$, $\hat{w} := w/\|w\|_2$, $w^\perp := \{\eta : \langle w,\eta\rangle = 0\}$, and $c, c' \in \mathbb{R}$,*

$$H_w(c) := \{\xi\in\mathbb{R}^d : \langle w,\xi\rangle = c\} = \frac{c}{\|w\|_2}\hat{w} + w^\perp, \qquad \operatorname{dist}\big(H_w(c),H_w(c')\big) = \frac{|c-c'|}{\|w\|_2}.$$

Proof.

$$\langle w,\xi\rangle = c \iff \xi - \frac{c}{\|w\|_2}\hat{w}\in w^\perp, \qquad \xi\in H_w(c),\ \eta\in H_w(c') \implies \|\xi-\eta\|_2 \ge \big|\langle\hat{w},\xi-\eta\rangle\big| = \frac{|c-c'|}{\|w\|_2},$$

$$\eta := \xi + \frac{c'-c}{\|w\|_2}\hat{w}\in H_w(c') \implies \|\xi-\eta\|_2 = \frac{|c-c'|}{\|w\|_2}.$$

□

Remark 2.1.5 (Linear Bradley–Terry models). Definition 1.1.2 is Definition 2.1.1 with $\theta_i = s_w(a_i)$, and Bradley–Terry on n items is the case $d = n$, $x(a_i) = e_i$, $w = \theta$,

$$\mathbb{P}(a_i \succ a_j \mid w) = \sigma\big(s_w(a_i) - s_w(a_j)\big) = \mathbb{P}_{(s_w(a_m))_m}(i \succ j), \qquad x(a_i) = e_i \implies \Delta_{ij}(w) = w_i - w_j.$$

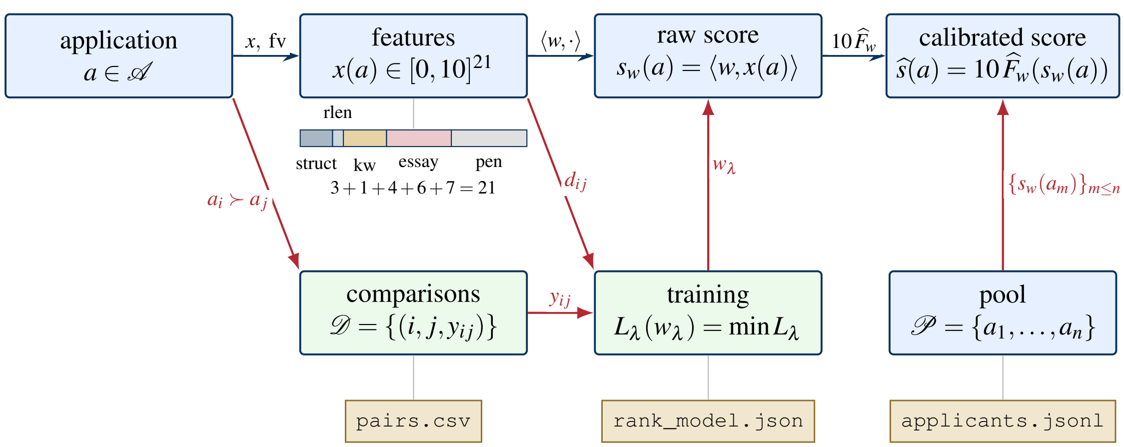

FIGURE 1. The pipeline. Applications are mapped deterministically to 21 bounded coordinates in five blocks, scored linearly, and calibrated against the pool, while human comparisons and feature differences train w_λ; the three files record labels, weights, and pool records.

3. FROM APPLICATIONS TO VECTORS

Every coordinate of the feature map is a bounded deterministic functional of one signal, so all randomness enters through the training of § 5 and none through extraction [5], [8, Chapter 3]. Throughout, Σ^* is the set of finite strings, $v \sqsubseteq T$ means that the phrase v occurs in T, $d = 21$, and an application a

carries its grade point average, calculus completion, entered course codes, browser-extracted resume text, and two short answers,

$$\mathrm{GPA}(a) \in [0, 4.33], \qquad \mathrm{Calc}(a) \in \{\mathrm{yes}, \mathrm{no}\}, \qquad U(a) \subset \Sigma^*, \qquad T(a), E_A(a), E_B(a) \in \Sigma^*.$$

3.1. **The five feature blocks.**

Definition 3.1.1 (Feature map and schema)**.** The deterministic feature map $x : \mathscr{A} \to [0,10]^{21}$ with schema identifier $\mathrm{fv} \in \mathbb{Z}_{\geq 1}$ concatenates the five blocks of Definitions 3.1.2–3.1.7,

$$x(a) = \big(x_{\mathrm{struct}}(a), x_{\mathrm{resume}}(a), x_{\mathrm{kw}}(a), x_{\mathrm{essay}}(a), x_{\mathrm{pen}}(a)\big) \in [0,10]^3 \times [0,10]^1 \times [0,10]^4 \times [0,10]^6 \times \{0,10\}^7.$$

Definition 3.1.2 (Structured coordinates)**.** For a whitelist $\mathscr{W}$ of course codes, the graduate codes $\mathscr{G}$ of the form MATH GR5xxx and MATH GR6xxx, and nondecreasing $\phi, \psi : [0,1] \to [0,1]$, for instance $\phi(u) = u^\alpha$ with $\alpha \in (0,1]$,

$$C(a) := U(a) \cap (\mathscr{W} \cup \mathscr{G}), \quad |C(a)| \leq 6, \qquad x_{\mathrm{gpa}}(a) := 10\,\phi\Big(\frac{\mathrm{GPA}(a)}{4.33}\Big), \qquad x_{\mathrm{upper}}(a) := 10\,\psi\Big(\frac{|C(a)|}{6}\Big),$$

$$x_{\mathrm{calc}}(a) := 10 \cdot \mathbf{1}\{\mathrm{Calc}(a) = \mathrm{yes}\}, \qquad x_{\mathrm{struct}}(a) := \big(x_{\mathrm{gpa}}, x_{\mathrm{calc}}, x_{\mathrm{upper}}\big)(a) \in [0,10] \times \{0,10\} \times [0,10].$$

Definition 3.1.3 (Resume length coordinate)**.** For $\ell(a) := |T(a)|$, $L_0 > 0$, and nondecreasing $s : [0,\infty) \to [0,1]$ with $s(0) = 0$, such as $\min\{1,u\}$ or $1 - e^{-u}$,

$$x_{\mathrm{resume}}(a) := x_{\mathrm{rlen}}(a) := 10\,s\Big(\frac{\ell(a)}{L_0}\Big) \in [0,10], \qquad \ell(a) = 0 \implies x_{\mathrm{rlen}}(a) = 0.$$

Definition 3.1.4 (Keyword coordinates)**.** For $\mathscr{K} := \{\mathrm{research}, \mathrm{teach}, \mathrm{leader}, \mathrm{awards}\}$ and, for each $c \in \mathscr{K}$, a finite dictionary $D_c \subset \Sigma^*$ and a threshold $\tau_c \geq 1$,

$$N_c(a) := \sum_{v \in D_c} \mathbf{1}\{v \sqsubseteq T(a)\}, \qquad x_c(a) := 10 \min\Big\{1, \frac{N_c(a)}{\tau_c}\Big\} \quad \Big(\text{or } 10\big(1 - e^{-N_c(a)/\tau_c}\big)\Big),$$

$$x_{\mathrm{kw}}(a) := \big(x_c(a)\big)_{c \in \mathscr{K}} \in [0,10]^4, \qquad \mathrm{KW}(a) := \frac{1}{4} \sum_{c \in \mathscr{K}} x_c(a) \in [0,10].$$

Example 3.1.5 (Keyword saturation)**.** For $\tau_{\mathrm{research}} = 3$, $\nu := N_{\mathrm{research}}$, $\chi := x_{\mathrm{research}}$, and resumes containing exactly REU, preprint, poster, or only REU,

$$\nu(a) = 3 \implies \chi(a) = 10, \qquad \nu(a) = 1 \implies \chi(a) = \tfrac{10}{3}, \qquad \nu(a) \geq 3 \implies \chi(a) = 10.$$

Definition 3.1.6 (Essay coordinates)**.** For word and sentence counts $\mathrm{wc}, \mathrm{sent} : \Sigma^* \to \mathbb{Z}_{\geq 0}$, word bounds $m \leq M$, reflection cues $R \subset \Sigma^*$ (realized, learned, mistake, revised, iterated, feedback, persist), action tokens $P \subset \Sigma^*$ (talk, problem session, reading group, outreach, workshop), a mathematical lexicon $\mathscr{M} \subset \Sigma^*$, thresholds $\rho, \varpi, \mu \geq 1$, and $\bullet \in \{A, B\}$,

$$x_{\mathrm{band},\bullet}(a) := 10 \cdot \mathbf{1}\{2 \leq \mathrm{sent}(E_\bullet(a)) \leq 4\} \quad \text{or} \quad x_{\mathrm{band},\bullet}(a) := 10 \cdot \mathbf{1}\{m \leq \mathrm{wc}(E_\bullet(a)) \leq M\},$$

$$x_{\mathrm{arc}}(a) := 10 \min\Big\{1, \frac{1}{\rho} \sum_{v \in R} \mathbf{1}\{v \sqsubseteq E_A(a)\}\Big\}, \qquad x_{\mathrm{plan}}(a) := 10 \min\Big\{1, \frac{1}{\varpi} \sum_{v \in P} \mathbf{1}\{v \sqsubseteq E_B(a)\}\Big\},$$

$$x_{\mathrm{math},\bullet}(a) := 10 \min\Big\{1, \frac{1}{\mu} \sum_{v \in \mathscr{M}} \mathbf{1}\{v \sqsubseteq E_\bullet(a)\}\Big\},$$

$$x_{\mathrm{essay}}(a) := \big(x_{\mathrm{band},A}, x_{\mathrm{band},B}, x_{\mathrm{arc}}, x_{\mathrm{plan}}, x_{\mathrm{math},A}, x_{\mathrm{math},B}\big)(a) \in [0,10]^6.$$

Definition 3.1.7 (Penalty block)**.** The block $x_{\mathrm{pen}}(a) \in \{0,10\}^7$ lists the indicators $10 \cdot \mathbf{1}\{\text{check fails}\}$ of seven validation checks, the sign of each influence on the rank being left to the learned weight.

Example 3.1.8 (Essay decomposition)**.** For a_1, a_2 agreeing in every coordinate except x_{plan}, with $E_B(a_1)$ naming at least ϖ actions of P and $E_B(a_2)$ none,

$$x_{\text{plan}}(a_1) = 10, \qquad x_{\text{plan}}(a_2) = 0, \qquad \Delta_{12}(w) = \langle w, d_{12} \rangle = 10\, w_{\text{plan}}, \qquad \mathbb{P}(a_1 \succ a_2 \mid w) = \sigma(10\, w_{\text{plan}}).$$

Proposition 3.1.9 (Boundedness and sensitivity of the feature map)**.** *For $a, b \in \mathscr{A}$, $w \in \mathbb{R}^d$, $c \in \mathscr{K}$, and $x'_c(a)$ the keyword coordinate after inserting one phrase into $T(a)$,*

$$\|x(a)\|_\infty \le 10, \qquad \|x(a) - x(b)\|_2^2 \le 100\, d, \qquad |s_w(a) - s_w(b)| \le 10\, \|w\|_1, \qquad 0 \le x'_c(a) - x_c(a) \le \frac{10}{\tau_c},$$

the last for both saturations of Definition 3.1.4.

Proof.

$$x(a), x(b) \in [0,10]^d \implies \|d_{ab}\|_\infty \le 10 \overset{2.1.3}{\implies} \|d_{ab}\|_2^2 \le 100\, d, \qquad |s_w(a) - s_w(b)| = |\langle w, d_{ab} \rangle| \le 10 \|w\|_1,$$

$$N_c(a) \le N'_c(a) \le N_c(a) + 1 \implies 0 \le \min\Big\{1, \frac{N'_c(a)}{\tau_c}\Big\} - \min\Big\{1, \frac{N_c(a)}{\tau_c}\Big\} \le \frac{1}{\tau_c},$$

$$0 \le e^{-N_c(a)/\tau_c} - e^{-N'_c(a)/\tau_c} \le 1 - e^{-1/\tau_c} \le \frac{1}{\tau_c}.$$

□

4. Linear scoring model

The score is linear so that each weight is the marginal effect of one named signal, and its geometry is that of projection onto $\hat{w} := w/\|w\|_2$ [5], [8, Chapter 3]. Throughout, $\xi \in \mathbb{R}^d$ denotes a feature vector and $H_w(c)$ the level sets of Lemma 2.1.4.

4.1. Scores, level sets, and scale.

Definition 4.1.1 (Score geometry)**.** For $w \in \mathbb{R}^d \setminus \{0\}$ and applications a, b,

$$s_w(a) = \|w\|_2\, \langle \hat{w}, x(a) \rangle, \qquad x(a) \in H_w\big(s_w(a)\big), \qquad \operatorname{dist}\big(H_w(s_w(a)), H_w(s_w(b))\big) \overset{2.1.4}{\implies} \frac{|\Delta_{ab}(w)|}{\|w\|_2},$$

so $x(a)$ and $x(b)$ lie on parallel hyperplanes whose separation along $\hat{w}$ is the margin divided by $\|w\|_2$.

Proposition 4.1.2 (Scale and order)**.** *For $w \in \mathbb{R}^d \setminus \{0\}$, $c > 0$, and applications a, b,*

$$s_{cw} = c\, s_w, \qquad \Delta_{ab}(cw) = c\, \Delta_{ab}(w), \qquad s_w(a) < s_w(b) \iff s_{cw}(a) < s_{cw}(b) \iff \langle \hat{w}, x(a) \rangle < \langle \hat{w}, x(b) \rangle,$$

$$\frac{\partial}{\partial \xi_k} \langle w, \xi \rangle = w_k, \qquad w \in [0,\infty)^d \implies 0 \le s_w(a) \le 10\, \|w\|_1,$$

so the ordering depends on $\hat{w}$ alone while the probabilities $\sigma(\Delta_{ab}(w))$ depend on $\|w\|_2$ as well.

Proof.

$$s_{cw}(a) = \langle cw, x(a) \rangle = c\, s_w(a) \implies \Delta_{ab}(cw) = c\Delta_{ab}(w), \quad c > 0 \implies s_w(a) < s_w(b) \iff s_{cw}(a) < s_{cw}(b),$$

$$c := \|w\|_2^{-1} \implies s_{cw}(a) = \langle \hat{w}, x(a) \rangle, \qquad \frac{\partial}{\partial \xi_k} \sum_{l=1}^{d} w_l \xi_l = w_k \overset{2.1.3}{\implies} 0 \le s_w(a) \le 10 \|w\|_1.$$

□

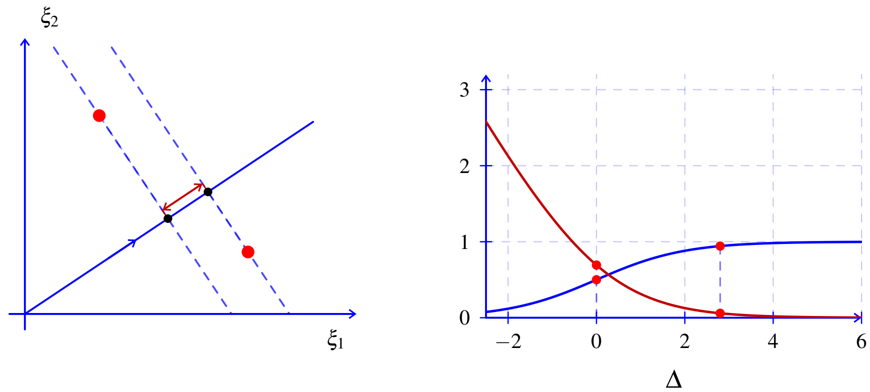

FIGURE 2. For $w = (3,2)$, $x(a) = (1.2,3.2)$, and $x(b) = (3.6,1)$, the left panel shows $H_w(10)$, $H_w(12.8)$, and separation $2.8/\sqrt{13}$; the right plots σ in blue and $\ell(\Delta) = \log(1+e^{-\Delta})$ in red, with $(\sigma,\ell)(0) = (\frac{1}{2},\log 2)$ and $(\sigma,\ell)(2.8) \approx (.943,.059)$.

5. Pairwise training objective

Maximum likelihood for the Bradley–Terry probabilities of Definition 1.1.2 is the minimization of a sum of logistic losses of signed margins, whose ℓ_2-regularized version is strongly convex and smooth with explicit constants, so that gradient descent converges linearly [1], [5], [8, Chapter 4], [9, Chapter 3]. Throughout, $\mathscr{D}$ is a pairwise dataset, $N := |\mathscr{D}|$, sums over (i,j,y) run over $\mathscr{D}$, σ' is evaluated at $\langle w, d_{ij}\rangle$ unless stated otherwise, and $\lambda \geq 0$.

5.1. Loss, gradient, convexity, and gradient descent.

Lemma 5.1.1 (Negative log-likelihood is the logistic loss). *For $(i,j,y) \in \mathscr{D}$, $z = 2y-1$, and $\Delta = \Delta_{ij}(w)$,*

$$-y\log\sigma(\Delta) - (1-y)\log\big(1-\sigma(\Delta)\big) = \log\big(1+e^{-z\Delta}\big) = \ell_{ij}(w),$$

$$\frac{1}{N}\sum_{(i,j,y)} \ell_{ij}(w) = -\frac{1}{N}\log\prod_{(i,j,y)} \mathbb{P}(a_i \succ a_j \mid w)^y\,\mathbb{P}(a_j \succ a_i \mid w)^{1-y}.$$

Proof.

$$y = 1 \implies z = 1, \quad -\log\sigma(\Delta) \overset{2.1.2}{\implies} \log\big(1+e^{-\Delta}\big) = \log\big(1+e^{-z\Delta}\big),$$

$$y = 0 \implies z = -1, \quad -\log\big(1-\sigma(\Delta)\big) \overset{2.1.2}{\implies} -\log\sigma(-\Delta) = \log\big(1+e^{\Delta}\big) = \log\big(1+e^{-z\Delta}\big),$$

$$\mathbb{P}(a_j \succ a_i \mid w) = \sigma\big(\Delta_{ji}(w)\big) = \sigma(-\Delta) \overset{2.1.2}{\implies} 1-\sigma(\Delta) \implies -\log\Big(\sigma(\Delta)^y\big(1-\sigma(\Delta)\big)^{1-y}\Big) = \ell_{ij}(w).$$

□

Lemma 5.1.2 (Gradient and Hessian). *For $w \in \mathbb{R}^d$,*

$$\nabla\ell_{ij}(w) = \big(\sigma(\Delta_{ij}(w)) - y_{ij}\big)\,d_{ij}, \qquad \nabla^2\ell_{ij}(w) = \sigma'\big(\Delta_{ij}(w)\big)\,d_{ij}\cdot d_{ij}^{\mathsf{T}},$$

$$\nabla L_\lambda(w) = \frac{1}{N}\sum_{(i,j,y)}\big(\sigma(\langle w,d_{ij}\rangle) - y\big)\,d_{ij} + \lambda w, \qquad \nabla^2 L_\lambda(w) = \frac{1}{N}\sum_{(i,j,y)}\sigma'\big(\langle w,d_{ij}\rangle\big)\,d_{ij}\cdot d_{ij}^{\mathsf{T}} + \lambda I.$$

Proof. Write $\Delta = \langle w,d\rangle$ with $d = d_{ij}$, $y = y_{ij}$, $z = 2y-1$, and differentiate $\ell_{ij}(w) = \log(1+e^{-z\Delta})$ by the chain rule.

$$\nabla_w\log\big(1+e^{-z\Delta}\big) = \frac{-z\,e^{-z\Delta}}{1+e^{-z\Delta}}\,d = -z\,\sigma(-z\Delta)\,d, \qquad \nabla_w\,\sigma(\langle w,d\rangle) = \sigma'(\Delta)\,d,$$

$$z=1 \implies -\sigma(-\Delta)\,d \overset{2.1.2}{\implies} \big(\sigma(\Delta)-1\big)\,d, \qquad z=-1 \implies \sigma(\Delta)\,d, \qquad \nabla\ell_{ij}(w)=\big(\sigma(\Delta)-y\big)\,d,$$
$$\nabla^2\ell_{ij}(w)=\sigma'(\Delta)\,d\cdot d^{\mathsf{T}}, \qquad \nabla\Big(\frac{\lambda}{2}\|w\|_2^2\Big)=\lambda w, \qquad \nabla^2\Big(\frac{\lambda}{2}\|w\|_2^2\Big)=\lambda I.$$

□

Proposition 5.1.3 (Convexity, strong convexity, and smoothness)**.** *For $\lambda \geq 0$, $\Lambda = \lambda + \frac{1}{4N}\sum_{(i,j,y)}\|d_{ij}\|_2^2$, and all $w \in \mathbb{R}^d$,*

$$\lambda I \preceq \nabla^2 L_\lambda(w) \preceq \Lambda I \preceq (\lambda+25d)\,I, \qquad L_\lambda \text{ convex},$$
$$L_\lambda \text{ strictly convex} \iff \lambda > 0 \text{ or } \operatorname{span}\{d_{ij} : (i,j,y) \in \mathscr{D}\} = \mathbb{R}^d,$$

and for $\lambda > 0$ there is a unique minimizer w_λ, characterized by its stationarity equation and bounded by

$$\lambda w_\lambda = \frac{1}{N}\sum_{(i,j,y)}\big(y-\sigma(\langle w_\lambda, d_{ij}\rangle)\big)\,d_{ij}, \qquad \|w_\lambda\|_2 \leq \frac{10\sqrt{d}}{\lambda}.$$

Proof.

$$0<\sigma' \leq \tfrac14 \overset{2.1.2,\,5.1.2}{\implies} \lambda I \preceq \nabla^2 L_\lambda(w) \preceq \lambda I + \frac{1}{4N}\sum_{(i,j,y)} d_{ij}\cdot d_{ij}^{\mathsf{T}} \preceq \Lambda I, \quad \|d_{ij}\|_2^2 \overset{3.1.9}{\implies} 100\,d \implies \Lambda \leq \lambda + 25d,$$

$$v^{\mathsf{T}}\cdot\nabla^2 L_0(w)\cdot v = \frac{1}{N}\sum_{(i,j,y)}\sigma'\big(\langle w, d_{ij}\rangle\big)\langle v, d_{ij}\rangle^2 = 0 \iff \langle v, d_{ij}\rangle = 0 \;\; \forall (i,j,y)\in\mathscr{D} \iff v \perp \operatorname{span}\{d_{ij}\},$$

$$\lambda > 0 \implies L_\lambda(w) \geq \frac{\lambda}{2}\|w\|_2^2 \longrightarrow \infty \quad (\|w\|_2\to\infty) \implies \exists!\, w_\lambda,$$

$$\nabla L_\lambda(w_\lambda)=0 \overset{5.1.2}{\iff} \lambda w_\lambda = \frac{1}{N}\sum_{(i,j,y)}\big(y-\sigma(\langle w_\lambda, d_{ij}\rangle)\big)\,d_{ij},$$

$$|y-\sigma| \leq 1, \quad \|d_{ij}\|_2 \leq 10\sqrt{d} \implies \lambda\|w_\lambda\|_2 \leq \frac{1}{N}\sum_{(i,j,y)}\|d_{ij}\|_2 \leq 10\sqrt{d}.$$

□

Theorem 5.1.4 (Gradient descent, [9])**.** *Let $\lambda > 0$, $w^{(0)} \in \mathbb{R}^d$, and $w^{(t+1)} := w^{(t)} - \Lambda^{-1}\nabla L_\lambda(w^{(t)})$. Then for every $t \geq 0$,*

$$L_\lambda(w^{(t)}) - L_\lambda(w_\lambda) \leq \Big(1-\frac{\lambda}{\Lambda}\Big)^t\big(L_\lambda(w^{(0)}) - L_\lambda(w_\lambda)\big),$$
$$\|w^{(t)} - w_\lambda\|_2^2 \leq \frac{2}{\lambda}\Big(1-\frac{\lambda}{\Lambda}\Big)^t\big(L_\lambda(w^{(0)}) - L_\lambda(w_\lambda)\big).$$

Proof. Write $f := L_\lambda$, $f^* := f(w_\lambda)$, $g := \nabla f(w^{(t)})$, and apply Lemma A.1.1 with $\mu = \lambda$ and $L = \Lambda$ through Proposition 5.1.3.

$$\nabla^2 f \preceq \Lambda I \overset{A.1.1}{\implies} f(w^{(t+1)}) \leq f(w^{(t)}) - \frac{1}{2\Lambda}\|g\|_2^2, \qquad \lambda I \preceq \nabla^2 f \overset{A.1.1}{\implies} \|g\|_2^2 \geq 2\lambda\big(f(w^{(t)}) - f^*\big),$$

$$f(w^{(t+1)}) - f^* \leq \Big(1-\frac{\lambda}{\Lambda}\Big)\big(f(w^{(t)}) - f^*\big) \implies f(w^{(t)}) - f^* \leq \Big(1-\frac{\lambda}{\Lambda}\Big)^t\big(f(w^{(0)}) - f^*\big),$$

$$\lambda I \preceq \nabla^2 f \overset{A.1.1}{\implies} \frac{\lambda}{2}\|w^{(t)} - w_\lambda\|_2^2 \leq f(w^{(t)}) - f^* \leq \Big(1-\frac{\lambda}{\Lambda}\Big)^t\big(f(w^{(0)}) - f^*\big).$$

□

Remark 5.1.5 (Optimization in practice). At the scale $d = 21$ and N between 10 and 10^3, a fixed number of epochs of the update $w^{(t+1)} = w^{(t)} - \eta \nabla L_\lambda(w^{(t)})$ with small λ suffices, each gradient costing $O(Nd)$ operations, since by Theorem 5.1.4

$$\Lambda \le \lambda + 25d = \lambda + 525, \qquad t \ge \frac{\Lambda}{\lambda} \log \frac{L_\lambda(w^{(0)}) - L_\lambda(w_\lambda)}{\varepsilon} \implies L_\lambda(w^{(t)}) - L_\lambda(w_\lambda) \le \varepsilon \quad (\varepsilon > 0).$$

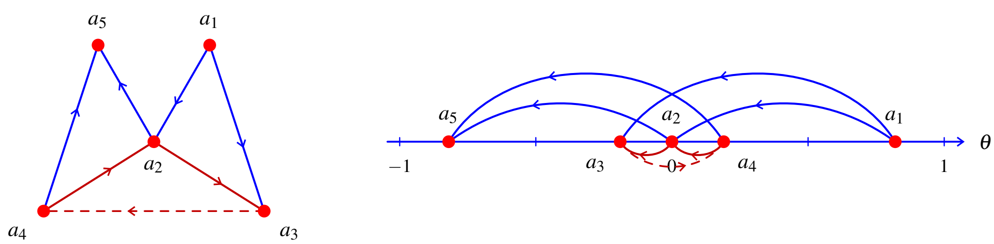

FIGURE 3. Seven comparisons on five applicants: the blue edges $a_1 \to a_2, a_1 \to a_3, a_4 \to a_5, a_2 \to a_5$ have probabilities $.694, .732, .732, .694$, the red cycle $a_2 \to a_3 \to a_4 \to a_2$ has probabilities $.546, .408, .546$ with its violated edge dashed, and the right panel places the same arrows at the $\lambda = .1$ fit $\theta = (.82, 0, -.19, .19, -.82)$.

6. CALIBRATION AND REPORTING

The objective of § 5 determines w through logit differences, so the raw scale of s_w is not intrinsic and applicant-facing reporting uses a monotone calibration to a bounded scale that preserves order [7], [8, Chapter 4]. Throughout, $\mathscr{P} = \{a_1, \dots, a_n\} \subset \mathscr{A}$ is the pool of a session or a reference set, e_c the coordinate vector of $c \in \mathscr{K}$, every submission stores the record $(s_w(a), \widehat{s}(a), \mathrm{KW}(a), x(a))$, and a session without a pool file logs $s_w(a)$ and sets the degenerate, non-comparable value $\widehat{s}(a) := 10$, so that the whole pipeline of § 3–§ 5 is the composite

$$\mathscr{A} \xrightarrow{x} [0,10]^{21} \xrightarrow{\langle w_\lambda, \cdot \rangle} \mathbb{R} \xrightarrow{10\widehat{F}_{w_\lambda}} \left\{ \frac{10}{n}, \frac{20}{n}, \dots, 10 \right\}.$$

6.1. Percentile calibration and decomposed reporting.

Definition 6.1.1 (Empirical distribution and calibrated score)**.** For $t \in \mathbb{R}$ and $a \in \mathscr{A}$,

$$\widehat{F}_w(t) := \frac{1}{n} \sum_{m=1}^{n} \mathbf{1}\{s_w(a_m) \le t\}, \qquad \widehat{s}(a) := 10\widehat{F}_w\big(s_w(a)\big) = \frac{10}{n} \big|\{m : s_w(a_m) \le s_w(a)\}\big| \in [0, 10].$$

Proposition 6.1.2 (Order preservation and invariance of the calibration)**.** *For $a = a_p$, $b = a_q$ in $\mathscr{P}$, $g : \mathbb{R} \to \mathbb{R}$ strictly increasing, $c > 0$, $\widehat{F}_{g \circ s_w}$ the empirical distribution of $\{g(s_w(a_m))\}_m$, and $\widehat{s}_w$ the calibrated score built from w,*

$$s_w(a) \gtreqless s_w(b) \iff \widehat{s}(a) \gtreqless \widehat{s}(b), \qquad \widehat{s}(\mathscr{P}) \subset \left\{ \frac{10}{n}, \dots, 10 \right\}, \qquad \max_{\mathscr{P}} \widehat{s} = 10,$$

$$\widehat{F}_{g \circ s_w} \circ g \circ s_w = \widehat{F}_w \circ s_w, \qquad \widehat{s}_{cw} = \widehat{s}_w.$$

Proof. Write $S_p := \{m : s_w(a_m) \le s_w(a_p)\}$, so that $\widehat{s}(a_p) = 10|S_p|/n$ and $p \in S_p$.

$$s_w(a_p) < s_w(a_q) \implies S_p \subsetneq S_q, \qquad s_w(a_p) = s_w(a_q) \implies S_p = S_q, \qquad s_w(a_p) \gtreqless s_w(a_q) \implies |S_p| \gtreqless |S_q|,$$

$$1 \le |S_p| \le n, \qquad s_w(a_p) = \max_m s_w(a_m) \implies S_p = \{1, \dots, n\}, \qquad \{m : g(s_w(a_m)) \le g(s_w(a_p))\} = S_p,$$

$$s_{cw} = c\, s_w \overset{4.1.2}{\implies} \widehat{F}_{cw}\big(s_{cw}(a_p)\big) = \widehat{F}_{c s_w}\big(c\, s_w(a_p)\big) = \widehat{F}_w\big(s_w(a_p)\big).$$

$\square$

Remark 6.1.3 (Keyword-only summary). The diagnostic KW of Definition 3.1.4 is the fixed linear functional

$$\mathrm{KW}(a) = \langle u, x(a)\rangle \in [0,10], \qquad u := \tfrac{1}{4}\sum_{c\in\mathscr{K}} e_c, \qquad s_w(a) - \mathrm{KW}(a) = \langle w-u, x(a)\rangle,$$

independent of the learned w and blind to the structured, essay, and penalty blocks, a reviewer-facing meter of resume evidence and not a replacement for s_w.

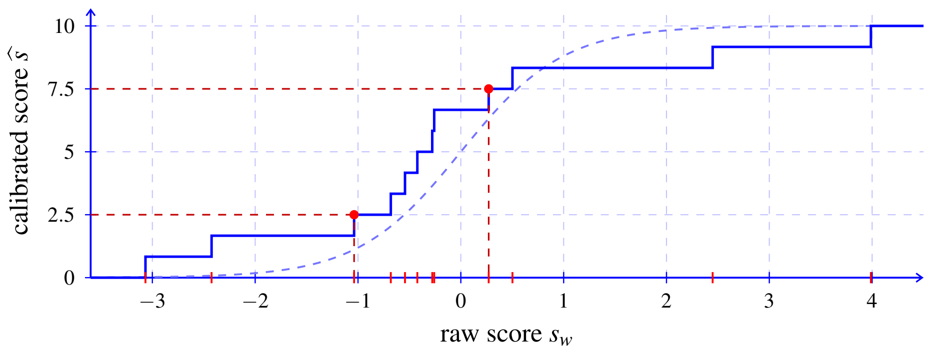

FIGURE 4. Percentile calibration for $n = 12$: the solid staircase $\widehat{s} = 10\widehat{F}_w(s_w)$ jumps by $10/12$ at the red score marks, the dashed curve is $10\sigma(2s)$, and the red guides give $\widehat{s}(b) = 2.5$ at $s_w(b) = -1.038$ and $\widehat{s}(a) = 7.5$ at $s_w(a) = .271$.

7. PROOF OF THEOREM A

Theorem 7.0.1 (Regularized pairwise logistic ranking)**.** *Let $x : \mathscr{A} \to [0,10]^d$ be a feature map, $\mathscr{D}$ a pairwise dataset, $\lambda > 0$, $\Lambda = \lambda + \frac{1}{4N}\sum_{(i,j,y)}\|d_{ij}\|_2^2$, and $\mathscr{P}$ a pool. Then*

(1) *$\lambda I \preceq \nabla^2 L_\lambda(w) \preceq \Lambda I \preceq (\lambda + 25d)I$ for all $w \in \mathbb{R}^d$, and L_λ has a unique minimizer w_λ with $\|w_\lambda\|_2 \le 10\sqrt{d}/\lambda$.*

Proof.

$$0 < \sigma' \le \tfrac{1}{4} \overset{2.1.2,\,3.1.9}{\Longrightarrow} \|d_{ij}\|_2^2 \le 100\,d \overset{5.1.3}{\Longrightarrow} \lambda I \preceq \nabla^2 L_\lambda \preceq \Lambda I \preceq (\lambda + 25d)\,I, \quad \exists!\, w_\lambda, \quad \|w_\lambda\|_2 \le \frac{10\sqrt{d}}{\lambda},$$

$\square$

(2) *w_λ is the unique solution of $\lambda w = \frac{1}{N}\sum_{(i,j,y)}\big(y - \sigma(\langle w, d_{ij}\rangle)\big)\, d_{ij}$.*

Proof.

$$\nabla L_\lambda(w) = 0 \overset{5.1.2}{\Longleftrightarrow} \lambda w = \frac{1}{N}\sum_{(i,j,y)}\big(y - \sigma(\langle w, d_{ij}\rangle)\big)\, d_{ij} \overset{5.1.3}{\Longleftrightarrow} w = w_\lambda,$$

$\square$

(3) *gradient descent with step Λ^{-1} satisfies $L_\lambda(w^{(t)}) - L_\lambda(w_\lambda) \le (1-\lambda/\Lambda)^t\big(L_\lambda(w^{(0)}) - L_\lambda(w_\lambda)\big)$.*

Proof.

$$w^{(t+1)} = w^{(t)} - \Lambda^{-1}\nabla L_\lambda(w^{(t)}) \overset{5.1.4}{\Longrightarrow} L_\lambda(w^{(t)}) - L_\lambda(w_\lambda) \le \Big(1 - \frac{\lambda}{\Lambda}\Big)^t \big(L_\lambda(w^{(0)}) - L_\lambda(w_\lambda)\big),$$

$\square$

(4) *for $a, b \in \mathscr{P}$, $s_w(a) \le s_w(b) \iff \widehat{s}(a) \le \widehat{s}(b)$, and $\widehat{s}$ is invariant under $w \mapsto cw$ for $c > 0$.*

Proof.

$$a,b\in\mathscr{P} \;\overset{6.1.2}{\Longrightarrow}\; \big(s_w(a)\le s_w(b) \iff \widehat{s}(a)\le\widehat{s}(b)\big), \qquad c>0 \;\overset{6.1.2}{\Longrightarrow}\; \widehat{s}_{cw}=\widehat{s}_w.$$

□

8. THE CONJECTURE

The regularization path $\lambda\mapsto w_\lambda$ is the one part of the pipeline whose qualitative behaviour is not settled by § 5–§ 6, and its large-λ limit is the anchor for what follows [8, Chapter 3], [7]. Throughout, $\mathscr{P}$ is a pool, $\mathscr{D}$ a pairwise dataset, $w^*\in\mathbb{R}^d$ a true weight vector, and

$$\bar d := \frac{1}{N}\sum_{(i,j,y)\in\mathscr{D}} \big(y_{ij}-\tfrac12\big)\,d_{ij}.$$

8.1. **The regularization path and its limit.**

Proposition 8.1.1 (Large regularization limit)**.** *For $\lambda>0$,*

$$\|\lambda w_\lambda-\bar d\|_2\le(\Lambda-\lambda)\|w_\lambda\|_2\le\frac{250\,d^{3/2}}{\lambda}, \qquad \lim_{\lambda\to\infty}\lambda\,w_\lambda=\bar d.$$

Proof.

$$\nabla L_\lambda(w_\lambda)=0 \;\overset{5.1.2}{\iff}\; \lambda w_\lambda-\bar d=\frac{1}{N}\sum_{(i,j,y)}\big(\tfrac12-\sigma(\langle w_\lambda,d_{ij}\rangle)\big)\,d_{ij}, \quad \sigma(0)=\tfrac12,\ \sigma'\le\tfrac14 \;\overset{2.1.2}{\Longrightarrow}\; \big|\tfrac12-\sigma(t)\big|\le\tfrac{|t|}{4},$$

$$\|\lambda w_\lambda-\bar d\|_2\le\frac{1}{4N}\sum_{(i,j,y)}|\langle w_\lambda,d_{ij}\rangle|\,\|d_{ij}\|_2\le\frac{\|w_\lambda\|_2}{4N}\sum_{(i,j,y)}\|d_{ij}\|_2^2=(\Lambda-\lambda)\|w_\lambda\|_2 \;\overset{3.1.9,\,5.1.3}{\Longrightarrow}\; \frac{25d\cdot10\sqrt{d}}{\lambda}.$$

□

Conjecture 8.1.2 (Finiteness and stabilization of the path)**.** *For $a,b\in\mathscr{P}$ with $\Delta_{ab}(w_\lambda)\not\equiv0$, and some $\lambda_0=\lambda_0(\mathscr{P},\mathscr{D})$,*

$$\big|\{\lambda>0:\Delta_{ab}(w_\lambda)=0\}\big|<\infty, \qquad \lambda\ge\lambda_0 \implies \big(\Delta_{ab}(w_\lambda)<0 \iff \langle\bar d,x(a)-x(b)\rangle<0\big).$$

Conjecture 8.1.3 (Sample complexity of the calibrated order)**.** *For labels drawn independently from Remark 2.1.5 with weights w^* on N pairs uniform in $\mathscr{P}$, $|\mathscr{P}|=n$, $\gamma>0$, and $\lambda\asymp N^{-1/2}$, there are $C,c>0$ with*

$$N\ge C\,d\,\gamma^{-2}\log n \implies \mathbb{P}(E_\gamma)\ge1-n^{-c}, \qquad E_\gamma:=\bigcap_{|\Delta_{ab}(w^*)|\ge\gamma}\big\{s_{w^*}(a)<s_{w^*}(b) \iff \widehat{s}(a)<\widehat{s}(b)\big\}.$$

Conjecture 8.1.4 (Calibration under pool drift)**.** *For pools $\mathscr{P}_0,\mathscr{P}_t$ of sizes n_0,n_t drawn from μ_0,μ_t on $\mathscr{A}$ with $W_1\big(\mu_0\circ s_w^{-1},\mu_t\circ s_w^{-1}\big)\le\eta$, and $\widehat{s}^{\mathscr{P}}$ the calibration of Definition 6.1.1 against $\mathscr{P}$, determine the least $\varepsilon=\varepsilon(\eta,n_0,n_t,\delta)$ and the reference size n_0 with*

$$\mathbb{P}\Big(\sup_{a\in\mathscr{A}}\big|\widehat{s}^{\mathscr{P}_t}(a)-\widehat{s}^{\mathscr{P}_0}(a)\big|\le\varepsilon\Big)\ge1-\delta,$$

replacing the degenerate solo value $\widehat{s}(a)=10$ of § 6 by a score comparable across sessions.

Part 2. Appendix

Appendix A. Foundatoional tools

A.1. Quadratic bounds and maximum likelihood.

Lemma A.1.1 (Quadratic bounds from Hessian bounds). *Let $f:\mathbb{R}^d\to\mathbb{R}$ be twice continuously differentiable with $\mu I \preceq \nabla^2 f(w) \preceq LI$ for all w, where $0<\mu\le L$, and let w_* be its minimizer. Then for all $v,w\in\mathbb{R}^d$,*

$$f(w)+\langle\nabla f(w),v-w\rangle+\frac{\mu}{2}\|v-w\|_2^2\le f(v)\le f(w)+\langle\nabla f(w),v-w\rangle+\frac{L}{2}\|v-w\|_2^2,$$
$$\frac{\mu}{2}\|w-w_*\|_2^2\le f(w)-f(w_*)\le\frac{1}{2\mu}\|\nabla f(w)\|_2^2,\qquad f\Big(w-\frac{1}{L}\nabla f(w)\Big)\le f(w)-\frac{1}{2L}\|\nabla f(w)\|_2^2.$$

Proof. Taylor's formula with integral remainder along the segment from w to v gives the first line, and the second follows by choosing v [9, Chapter 3].

$$f(v)=f(w)+\langle\nabla f(w),v-w\rangle+\int_0^1(1-\theta)\,(v-w)^{\mathsf T}\cdot\nabla^2 f\big(w+\theta(v-w)\big)\cdot(v-w)\,d\theta,$$
$$\int_0^1(1-\theta)\,d\theta=\frac{1}{2},\qquad \mu\|v-w\|_2^2\le(v-w)^{\mathsf T}\cdot\nabla^2 f\big(w+\theta(v-w)\big)\cdot(v-w)\le L\|v-w\|_2^2,$$
$$v:=w-\frac{1}{L}\nabla f(w)\implies f(v)\le f(w)-\frac{1}{L}\|\nabla f(w)\|_2^2+\frac{L}{2}\cdot\frac{1}{L^2}\|\nabla f(w)\|_2^2=f(w)-\frac{1}{2L}\|\nabla f(w)\|_2^2,$$
$$v:=w_*\implies f(w_*)\ge f(w)+\min_{u\in\mathbb{R}^d}\Big(\langle\nabla f(w),u\rangle+\frac{\mu}{2}\|u\|_2^2\Big)=f(w)-\frac{1}{2\mu}\|\nabla f(w)\|_2^2,$$
$$w:=w_*,\ \nabla f(w_*)=0\implies f(v)\ge f(w_*)+\frac{\mu}{2}\|v-w_*\|_2^2\quad(v\in\mathbb{R}^d).$$

□

Remark A.1.2 (Maximum likelihood and ridge penalty). For conditionally independent labels with $\mathbb{P}(y_{ij}=1\mid w)=\sigma(\Delta_{ij}(w))$ and the Gaussian prior $w\sim\mathcal{N}(0,(N\lambda)^{-1}I)$,

$$-\frac{1}{N}\log\prod_{(i,j,y)}\sigma\big(\Delta_{ij}(w)\big)^y\big(1-\sigma(\Delta_{ij}(w))\big)^{1-y}\overset{5.1.1}{\implies}\frac{1}{N}\sum_{(i,j,y)}\ell_{ij}(w),$$
$$-\frac{1}{N}\log\exp\Big(-\frac{N\lambda}{2}\|w\|_2^2\Big)=\frac{\lambda}{2}\|w\|_2^2,\qquad L_\lambda(w)=\frac{1}{N}\sum_{(i,j,y)}\ell_{ij}(w)+\frac{\lambda}{2}\|w\|_2^2,$$

so w_λ is the maximum-a-posteriori estimate, and w_0, when $\operatorname{span}\{d_{ij}\}=\mathbb{R}^d$ and the minimum of L_0 is attained, is the maximum-likelihood estimate [8, Chapters 3–4].

Appendix B. Data artifacts

Three files connect deployed scoring to local training, and the determinism of Definition 3.1.1 is what makes a score computed in the browser reproducible from the stored records.

B.1. Records, pairs, and the model file.

Definition B.1.1 (Artifacts). For a pool $\mathscr{P}=\{a_1,\dots,a_n\}$, a pairwise dataset $\mathscr{D}$, and a weight vector w_λ trained on the schema fv,

$$\texttt{applicants.jsonl}:=\big\{\big(m,\ \mathrm{fv},\ x(a_m),\ s_{w_\lambda}(a_m),\ \widehat{s}(a_m),\ \mathrm{KW}(a_m)\big)\big\}_{m=1}^n,$$
$$\texttt{pairs.csv}:=\big\{(i,j,y_{ij})\big\}_{(i,j,y)\in\mathscr{D}},\qquad \texttt{rank\_model.json}:=\big(\mathrm{fv},\ \lambda,\ w_\lambda\big),$$

one record per applicant, one row per comparison, and one weight vector with the schema it was trained on.

Proposition B.1.2 (Reproducibility of deployed scores)**.** *If the schema of* `rank_model.json` *equals the schema of the deployed feature map, then for every $a \in \mathscr{P}$ the raw score recomputed at deployment equals the stored one, and the calibrated score and the keyword summary are functions of the stored records alone,*

$$\mathrm{fv}_{\mathrm{model}} = \mathrm{fv}_{\mathrm{deploy}} \implies \langle w_\lambda, x(a)\rangle = s_{w_\lambda}(a), \qquad \mathrm{KW}(a) = \langle u, x(a)\rangle,$$

$$\widehat{s}(a) = \frac{10}{n}\big|\{m : s_{w_\lambda}(a_m) \le s_{w_\lambda}(a)\}\big|.$$

Proof. Definition 3.1.1, Definition 6.1.1, Remark 6.1.3. □

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